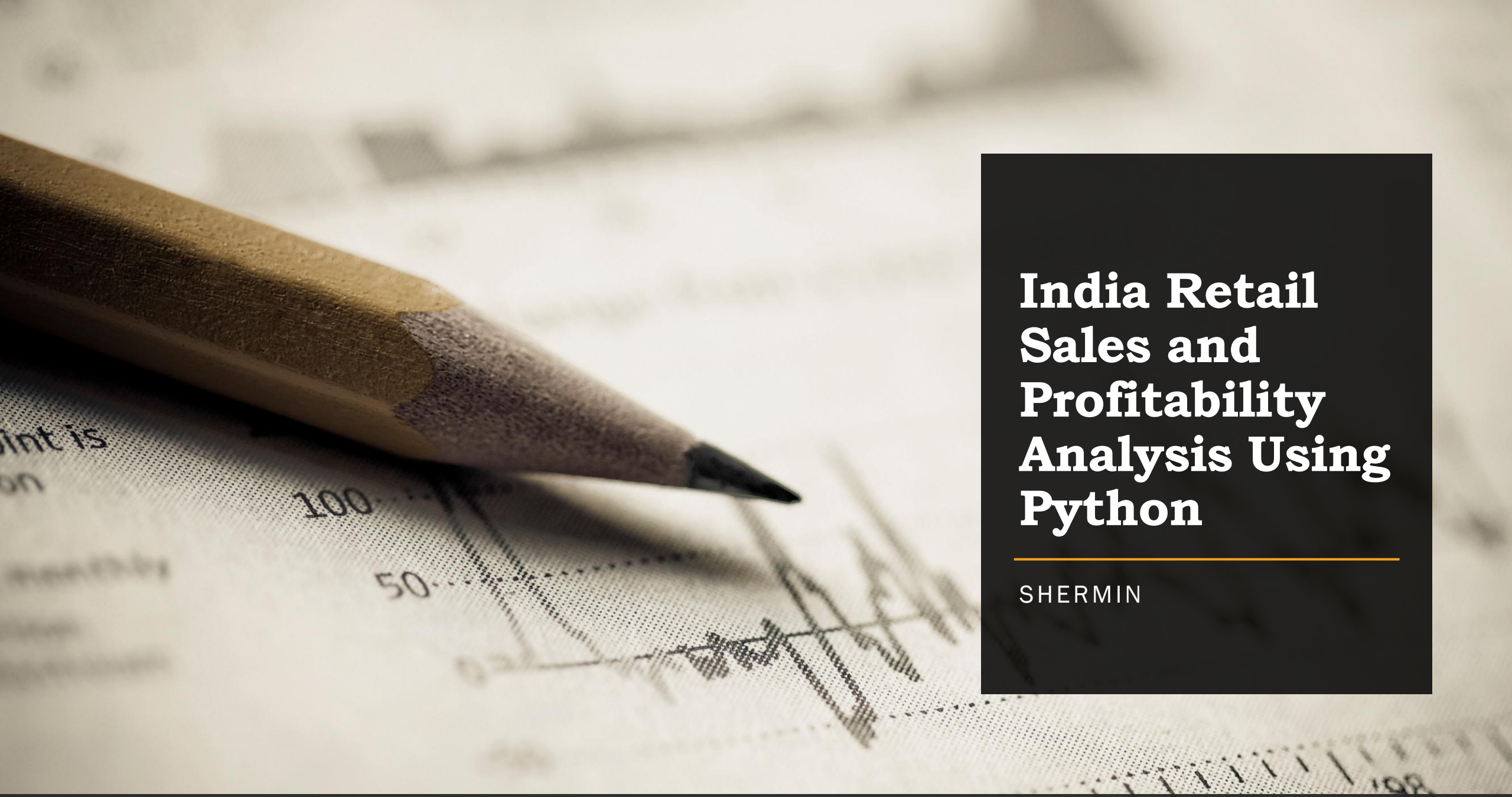


India Retail Sales and Profitability Analysis Using Python

SHERMIN

Problem Statement, Objectives & Business Questions

Problem Statement

Analyze Indian retail sales data to understand **sales performance, profitability, product/segment/region performance, and factors associated with profit and loss.**

Objectives

- Examine the structure and quality of the retail dataset
- Analyze sales, profit, quantity, discount and freight expenses
- Compare performance across **regions, states, segments and product types**
- Identify patterns in **profitable and loss-making orders**
- Use statistical and visual analysis to generate business insights

Key Business Questions

- How do sales and profitability vary across regions, states, products, and customer segments?
- What factors are associated with sales and profit, including quantity, discounts, and freight expenses?
- How do profitable and loss-making orders differ in their sales and cost patterns?
- How do combinations of region, segment, product type, and order priority affect profitability?
- How does business performance change over time, and where are unusual patterns or potential outliers found?

Dataset Overview & Initial Observation

Attribute	Details
Records	2,594 rows
Variables	18 columns
Domain	Retail / Sales
Data type	Numerical + Categorical + Date
Data quality	No missing values
Duplicate records	None
Numerical validation	No invalid/negative values identified

Initial Observation

The dataset contains reliable and useful information for analyzing retail sales and profitability across different business dimensions.

Main variables include :-

Sales • Profit • Quantity • Discount • Freight Expenses • Region • State • Segment • Product Type • Order Priority • Dates

Data Preprocessing

Data Preprocessing

1. Data Quality Checks
- Missing-value analysis
- Duplicate-record check
- Data-type validation
- Numerical-value validation

2. Derived Features Created

Derived Column	Purpose
shipping_days	Measure delivery duration
profit_margin	Measure profitability relative to sales
total_cost	Estimate total cost associated with an order
sales_per_unit	Compare sales efficiency per quantity
discount_percentage	Express discount as a percentage

Data Filtering & Aggregation

Filtered data tables:-
1) Profitable Orders: Profit > 0
2) Loss Orders: Profit < 0
Additional aggregation was performed using GroupBy to compare total profit across product types.

Exploratory Data Analysis

1. Univariate Analysis

Understanding the distribution and structure of individual **numerical and categorical variables**.

2. Bivariate Analysis

Examining relationships between two variables, including:

- Sales vs. Profit
- Profit vs. Quantity
- Profit vs. Freight Expenses
- Profit vs. Discount

Analysis was performed on:
Full Dataset | Profitable Orders | Loss Orders

3. GroupBy Analysis

Comparing:

- Sales and profit by region
- Total and average profit by product type
- Segment performance
- Performance by region
- Performance by state

4. Pivot Table Analysis

Used for multi-dimensional comparisons involving:

- Region
- Segment
- Product Type
- Order Priority
- Sales / Profit / Freight Expenses

5. Multivariate Analysis

Examined combinations of Region + Segment + Product Type and Region + Product Type + Freight Mode.

6. Correlation Analysis

Measured relationships among numerical variables to identify linear associations.

Data Visualization

Analysis	Visualization	Python Library
Univariate Analysis	Bar Chart – Orders by Region	Matplotlib
	Pie Chart – Orders by Segment	Matplotlib
Bivariate Analysis	PairGrid – Relationships between Numerical Variables	Seaborn
GroupBy Analysis	Bar Chart – Sales by Region	Matplotlib
	Bar chart – Profit by Product Type	Seaborn
	Bar Chart – Average Profit by Product Type	Matplotlib
	Violin Plot – Profit Distribution by Product Type	Seaborn
	Interactive Bar Chart – Sales & Profit by Segment	Plotly
	Pie Chart – Profit Margin by Segment	Plotly
	Bar chart – Sales & Profit by Region	Seaborn
	Pie Chart – Profit Margin by Region	Plotly
	Horizontal Bar Chart – Profit Margin by State	Matplotlib
	Box Plot – State Profit Margin / Outlier Identification	Seaborn
	Box Plot – Profit Distribution by Segment	Seaborn

Data Visualization

Analysis	Visualization	Python Library
Pivot Table Analysis	Heatmap – Sales by Region & Segment	Seaborn
	Heatmap – Profit by Region & Segment	Seaborn
	Line Chart – Average Profit by Order Priority & Segment	Matplotlib
Multivariate Analysis	Bar Chart – Profit by Region, Segment & Product Type	Matplotlib
	Interactive Treemap – Freight Expenses by Region, Product Type & Freight Mode	Plotly
Correlation Analysis	Correlation Heatmap	Seaborn
Time & Distribution Analysis	Line Chart – Monthly Sales Trend	Matplotlib
	Histogram – Discount Offered	Matplotlib
	Interactive Line Chart – Monthly Profit Trend	Plotly

Key Business Insights

1. Regional sales performance
The North region recorded the highest total sales, followed closely by the West region, while the East region recorded the lowest sales. This shows that sales performance varies considerably across regions, indicating differences in regional market contribution.

2. Product profitability
Processed Meat generated the highest total profit, followed by Canned Foods and Preserved Food. However, when average profit per order is considered, Canned Foods had the highest average profit per order. This shows that total profitability and average profitability can provide different perspectives on product performance.

3. Customer segment profitability
The Stand Alone Restaurants segment recorded the highest profit margin at approximately 20.15%, while the Personel Usage segment had the lowest profit margin at approximately 6.44%. This indicates that some customer segments generate more profit relative to their sales than others.

4. Sales and profit relationship
The correlation analysis showed a weak positive relationship between Sales and Profit (approximately 0.27). This indicates that higher sales are generally associated with higher profit, but the relationship is not strong. Therefore, increasing sales alone does not necessarily guarantee higher profitability.

Key Business Insights

5. Multiple factors affect profitability
Profit showed only weak or very weak relationships with variables such as quantity ordered, discount offered, unit price, and freight expenses. This suggests that profitability is influenced by multiple factors rather than being strongly associated with one single numerical variable.

6. Difference between profitable and loss-making orders
The analysis of profitable and loss-making orders showed that their relationships between sales, profit, quantity, and freight expenses differ. In particular, some loss-making orders have relatively high sales but still generate negative profit. This indicates that high sales do not always translate into profitable orders, and costs such as freight expenses and other order-level factors need to be considered.

7. State-level profitability and outliers
State-level analysis revealed substantial differences in profit margins. Himachal Pradesh showed an unusually low profit margin, which was strongly influenced by an extreme loss-making order in the Processed Meat category. This order was treated as a potential outlier rather than being removed because there was insufficient evidence to confirm that it was an incorrect record.

8. Region and segment performance
The pivot table analysis showed that sales and profit vary across combinations of region and customer segment. For example, the Personal Usage segment in the North region recorded the highest total profit in the region–segment profit comparison, while the Personal Usage segment in the South region recorded the lowest.

Key Business Insights

9. Order priority and average profit

The **analysis** of order priority and customer segment showed that average profit varies across different combinations. Stand Alone Restaurants recorded their highest average profit at the Low order-priority level, while Personal Usage recorded the lowest average profit at the Not Specified priority level. This shows that average profitability differs across both customer segments and order-priority categories.

10. Time-based performance

The monthly trend analysis showed noticeable variation in profitability over time. November 2012 recorded the highest monthly profit at approximately 27.98K, while May 2012 recorded the lowest monthly profit at approximately −10.22K. This indicates that business profitability was not consistent across all months and that certain periods performed considerably better or worse than others.

11. Freight expense patterns

The multivariate analysis showed that freight expenses vary across regions, product types, and freight modes. This indicates that shipping-related costs are not uniform across all orders and that certain combinations of region, product type, and freight mode may require closer monitoring.

12. Profit distribution and unusual orders

The box plot and violin plot showed considerable variation in individual order-level profits, including several outliers across customer segments. The violin plots also showed that many product-type orders have profit values concentrated around zero, while a smaller number of orders have much higher profits or losses. This highlights the importance of examining individual orders in addition to aggregated business performance.

Conclusion

The analysis reveals that profitability depends on more than sales volume.

North and West are the leading regions by sales.

Processed Meat contributes the highest total profit.

Canned Foods achieves the highest average profit per order.

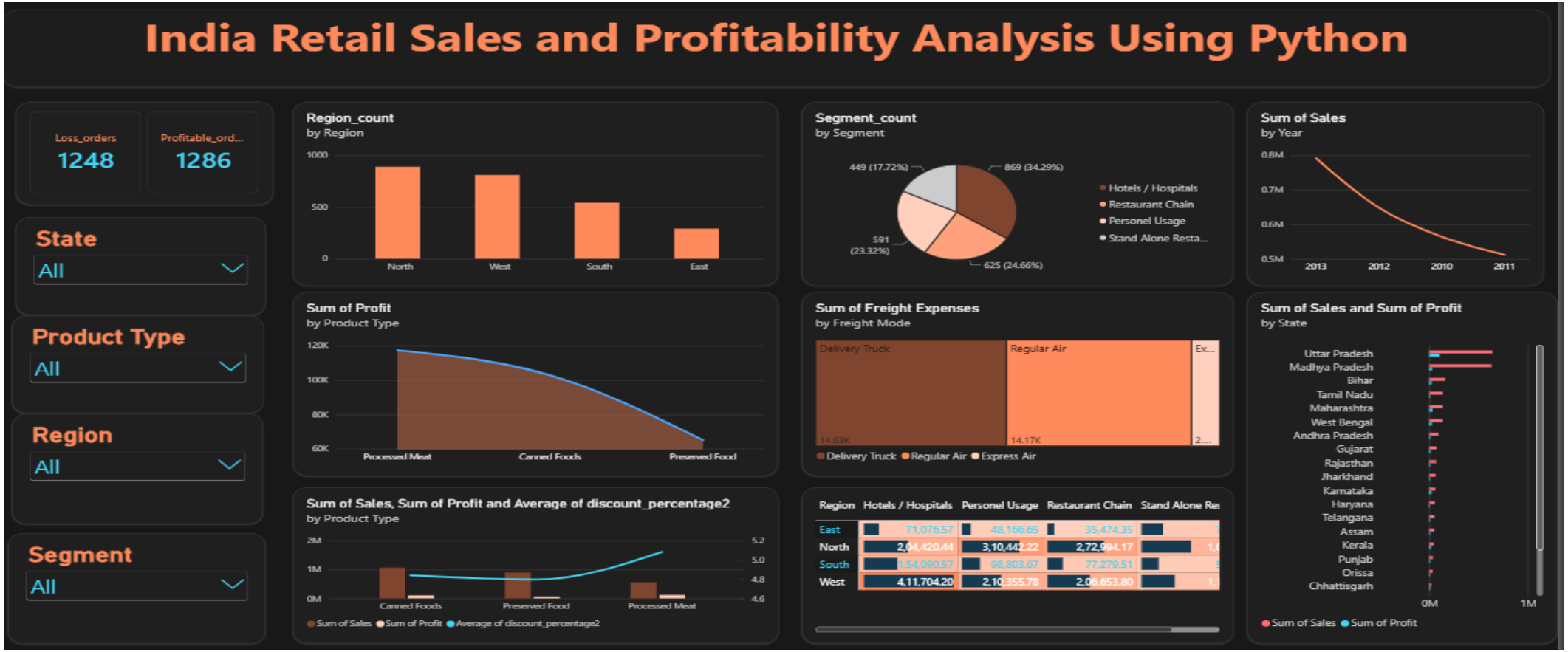
Stand Alone Restaurants show the highest average profit margin.

Discounts and quantity have only weak relationships with profit.

Outlier transactions can significantly affect profitability analysis.

Overall, the analysis provides data-driven insights into regional, product, and customer-segment performance and highlights opportunities for improving profitability.

Dashboard


India Retail Sales and Profitability Analysis Using Python
Loss_orders
1248
Profitable_ord...
1286
State
All
Product Type
All
Region
All
Segment
All
Region_count
by Region
1000
500
0
North
West
South
East
Segment_count
by Segment
449 (17.72%)
869 (34.29%)
591 (23.32%)
625 (24.66%)
Hotels / Hospitals
Restaurant Chain
Personel Usage
Stand Alone Resta...
Sum of Sales
by Year
0.8M
0.7M
0.6M
0.5M
2013
2012
2010
2011
Sum of Profit
by Product Type
120K
100K
80K
60K
Processed Meat
Canned Foods
Preserved Food
Sum of Freight Expenses
by Freight Mode
Delivery Truck
Regular Air
14.63K
14.17K
Delivery Truck
Regular Air
Express Air
Sum of Sales and Sum of Profit
by State
Uttar Pradesh
Madhya Pradesh
Bihar
Tamil Nadu
Maharashtra
West Bengal
Andhra Pradesh
Gujarat
Rajasthan
Jharkhand
Karnataka
Haryana
Telangana
Assam
Kerala
Punjab
Orissa
Chhattisgarh
0M
1M
Sum of Sales
Sum of Profit
Sum of Sales, Sum of Profit and Average of discount_percentage2
by Product Type
2M
1M
0M
5.2
5.0
4.8
4.6
Canned Foods
Preserved Food
Processed Meat
Sum of Sales
Sum of Profit
Average of discount_percentage2
Region
Hotels / Hospitals
Personel Usage
Restaurant Chain
East
71,076.57
48,166.65
35,474.35
North
2,04,420.44
3,10,442.22
2,72,994.17
South
1,54,090.57
98,803.67
77,279.51
West
4,11,704.20
2,10,355.78
2,06,653.80

Thank You